

Customer Churn Analysis

Project Report

Prepared from the churn analysis notebook and SQLite customer dataset

Analysis date: 16 July 2026

1. Executive Summary

This project integrates customer, subscription, and support records to identify customer attrition patterns. The analysis covers 21 unique customers, of whom 6 cancelled their subscription. The observed churn rate is 28.57%, indicating that retention should be prioritized, particularly for the Basic plan and customers showing high churn-risk scores.

KPI	Result	Business interpretation
Churn rate	28.57%	6 of 21 customers cancelled.
Retention rate	71.43%	Customers remaining active in the observation set.
ARPU	Rs 18.85	Average monthly recurring revenue per customer.
Revenue at risk	Rs 73.94	Monthly charges associated with churned customers.
Average tenure	1496 days	Mean service duration, using cancellation date or analysis date.
Escalation rate	19.05%	Share of customers with a latest support escalation.
Avg. complaints per user	0.43	Complaint events normalized by unique customers.

2. Data Sources and Data Preparation

The source database contains three relational tables linked through customerid. A left join preserves the subscription population while enriching each record with customer demographics and the latest available support signal.

Table	Key fields used	Transformation
db_customer	customerid, country, state, gender, dob	Renamed name to customer_name; removed sparse interests and empty pincode; standardized Men/Women to Male/Female; converted DOB to date; filled country from geographic context.
db_subscription	dates, plan/contract, cancellation, charges, CLTV, churn score	Converted date fields; derived churn_flag = 1 when cancellation_date exists; derived tenure_days and churn_risk.
db_support	complaint date, escalations, CSAT	Dropped unused columns; counted complaints; retained the latest complaint per customer to avoid duplicate joins; encoded escalation Y/N as 1/0.

3. KPI Definitions

- Churn Rate: mean of churn_flag multiplied by 100; cancellation_date is the churn event.
- Retention Rate: 100 minus churn rate.
- ARPU (Average Revenue Per User): mean monthly_charges across customers.
- Revenue at Risk: sum of monthly_charges for churned customers; it is a monthly exposure proxy, not lifetime value loss.
- Customer Tenure: days between subscription start and cancellation; active customers are measured through 16 July 2026.
- Churn Risk Band: Low (<50), Medium (50-69), High (>=70) based on churn_score.

4. Pivot Tables and Segment Analysis

The primary pivot table summarizes plan-level customer volume, monthly charges, and average churn. It directly identifies the Basic plan as the highest-risk product segment.

Plan type	Churn rate	Customers	Monthly charges (Rs)
Basic	60.00%	5	52.95
Standard	22.22%	9	123.91
Premium	14.29%	7	218.93

Key finding: Basic customers churn at 60.00%, versus 22.22% for Standard and 14.29% for Premium. This is a small sample, so results should be validated on a larger extract before setting policy.

5. Charts Created

The following charts were created in the notebook and reproduced for this report. Percent-based charts use churn_flag averages; the monthly trend counts cancellations by month.

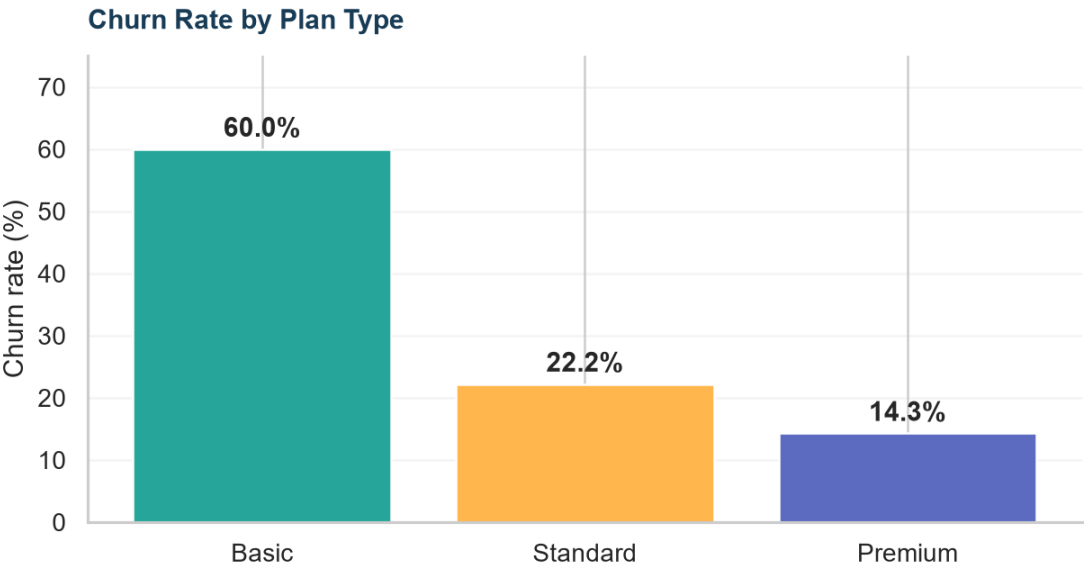


Figure 1. Churn rate by plan type.

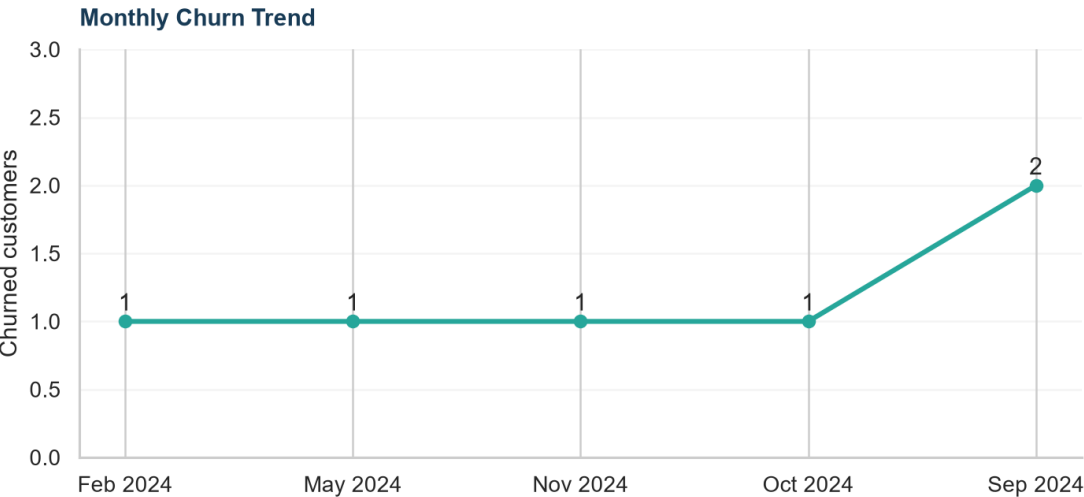


Figure 2. Monthly churn trend based on cancellation month.

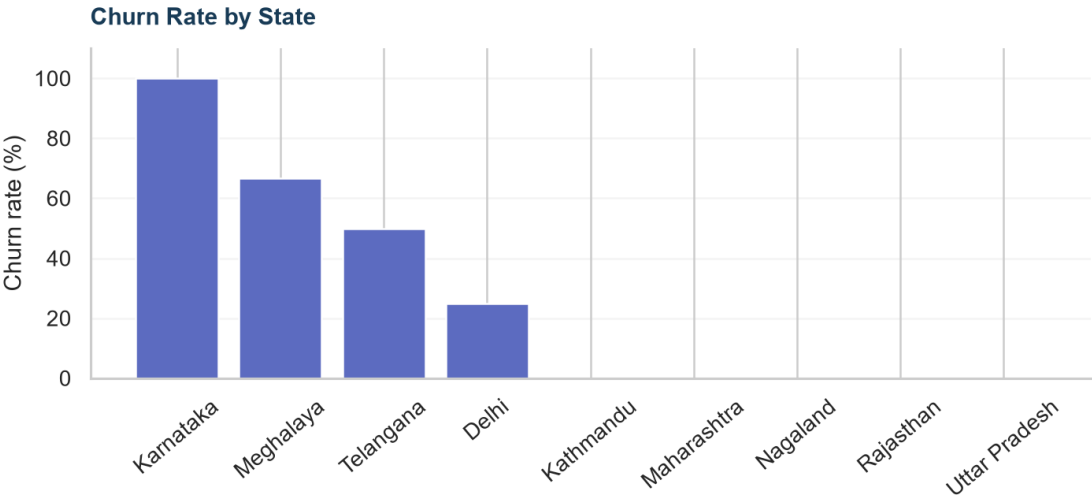


Figure 3. Churn rate by state.

6. Correlation and Diagnostic Visuals

A correlation heatmap was produced after ordinal encoding: Basic/Standard/Premium for plan tier, Monthly/Annual for contract type, and Low/Medium/High for risk band. Categorical-code correlations are exploratory only; they should not be interpreted as causal effects.

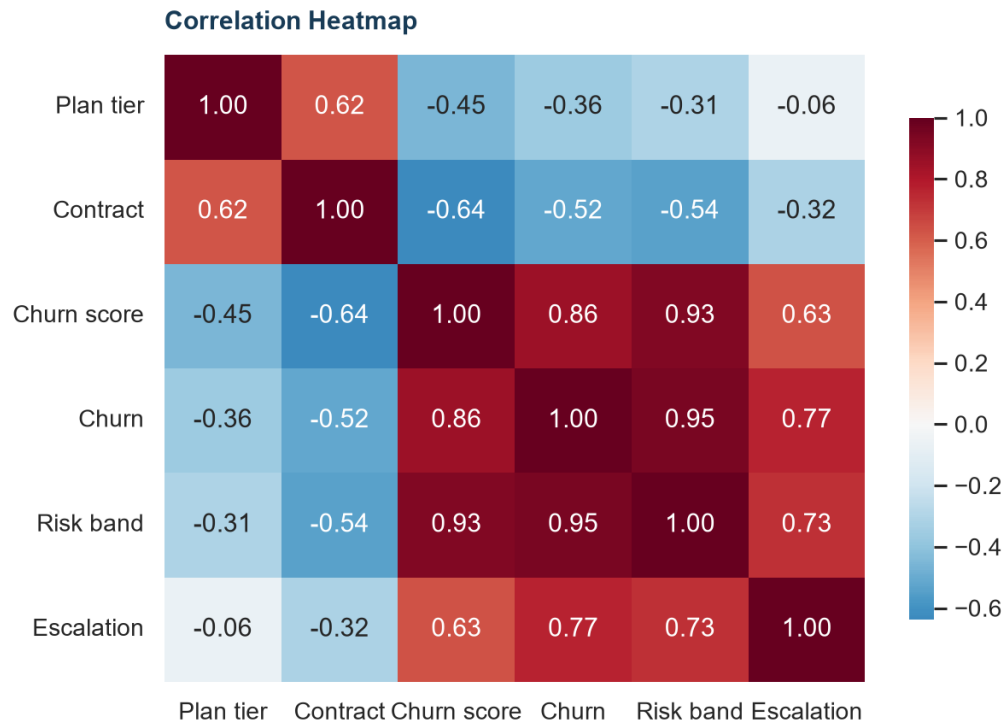


Figure 4. Correlation heatmap for encoded churn-related features.

7. Findings and Recommendations

- Prioritize the Basic plan: its 60.00% churn rate is materially higher than the other plan segments.
- Use churn score for outreach: 6 customers fall into the High-risk band; combine this signal with billing, product, and support context before intervening.
- Escalations show an exploratory correlation of 0.77 with churn in the prepared dataset. Treat it as a flag for service-recovery investigation, not a causal conclusion.
- Create a retention playbook: targeted renewal reminders, price/value messaging for cancellation-prone cohorts, and fast follow-up after escalated complaints.
- Operationalize a monthly KPI dashboard tracking churn rate, retention, ARPU, revenue at risk, tenure, escalation rate, and plan-level churn.

8. Limitations and Next Steps

The dataset is limited to 21 customers and seven deduplicated support records, so percentages may shift substantially with a larger sample. Country imputation and support record deduplication are documented assumptions. Future iterations should include cohort-based churn, survival/tenure modeling, cancellation reason coding, CSAT trend analysis, and a validated predictive model with a held-out test set.

Appendix: Technical Terms

Term	Meaning
Churn	Customer cancellation or loss of service.
CLTV	Customer Lifetime Value; the value attributed to a customer over the relationship.
ARPU	Average Revenue Per User, here based on monthly charges.
KPI	Key Performance Indicator used to monitor operational or business health.
Pivot table	A grouped aggregation used to compare measures across categories.
Correlation	A measure of linear association; it does not establish causation.
Left join	A merge that keeps every record from the primary (subscription) table.